NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - II

Interpolation & Curve Fitting

Syllabus: Finite differences - Newton's forward and backward interpolation formulae - Lagrange's formula. Curve fitting: Fitting of straight line, second-degree and Exponential curve by method of least squares.

I. Introduction to Interpolation

Interpolation is the process of finding the value of a function $y = f(x)$ for a value of x lying **inside** the range of given data (x_0, y_0), (x_1, y_1), ... (x_n, y_n). Finding a value outside the range is called **extrapolation**.

We replace the unknown function by a simple polynomial $P(x)$ of degree $\leq n$ which passes through all the $(n + 1)$ given points : $P(x_i) = y_i$, $i = 0, 1, \ldots n$.

- **Equal spacing** of x ($x_1 - x_0 = h$) : Newton's forward / backward formulae.
- **Unequal spacing** : Lagrange's formula.

2. Finite Differences

Let the values of x be equally spaced with step h : $x_i = x_0 + ih$, and $y_i = f(x_i)$.

2.1 Forward difference operator Δ

$$\Delta y_i = y_{i+1} - y_i \text{ (first forward difference)}$$

$$\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i \ , \ \Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$$

2.2 Backward difference operator ∇

$$\nabla y_i = y_i - y_{i-1} \ , \ \nabla^2 y_i = \nabla y_i - \nabla y_{i-1}$$

2.3 Other operators

- **Shift operator E** : $E\, f(x) = f(x + h)$, so $E\, y_i = y_{i+1}$ and $E^n\, y_i = y_{i+n}$.
- **Central difference δ** : $\delta\, y_i = y(i+½) - y(i-½)$.
- **Averaging μ** : $\mu\, y_i = ½\, [\, y(i+½) + y(i-½)\,]$.

Relations : $\Delta = E - 1$, $\nabla = 1 - E^{-1}$, $E = 1 + \Delta$, $\Delta\nabla = \nabla\Delta = \delta^2$, $\nabla = \Delta E^{-1}$

Important results

- The nth difference of a polynomial of degree n is **constant** ($= n!\, h^n\, a_n$) and the (n+1)th difference is zero.
- $\Delta(c) = 0$; $\Delta(e^{ax}) = (e^{ah} - 1)\, e^{ax}$; $\Delta(f + g) = \Delta f + \Delta g$.

2.4 Forward difference table (n = 4 points shown)

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$
x_1	y_1	Δy_1	$\Delta^2 y_1$	$\Delta^3 y_1$	
x_2	y_2	Δy_2	$\Delta^2 y_2$		
x_3	y_3	Δy_3			
x_4	y_4				

In backward table the same entries are written along the **last row** :

∇y_n, $\nabla^2 y_n$, $\nabla^3 y_n$ ...

Example 1 : difference table for $y = x^3$

x	$y = x^3$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	1	7	12	6	0
2	8	19	18	6	
3	27	37	24		
4	64	61			
5	125				

Third difference is constant ($6 = 3!$) so the data come from a cubic polynomial.

Missing term problem

If one value is missing among n known values, assume the (n)th differences are zero, i.e. $\Delta^n y_0 = 0$; expand in terms of E : $(E - 1)^n y_0 = 0$.

Example : 4 values known, 1 missing → use $\Delta^4 y_0 = 0$ → $y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$.

3. Newton's Forward Interpolation Formula

Used when the required x is **near the beginning** of the table. Put **$p = (x - x_0)/h$**.

$$y(x) = y_0 + p\Delta y_0 + [p(p-1)/2!]\,\Delta^2 y_0 + [p(p-1)(p-2)/3!]\,\Delta^3 y_0 + \dots$$

Derivation idea : $y(x_0 + ph) = E^p\, y_0 = (1 + \Delta)^p\, y_0$ and expand by binomial theorem.

Example 2

From the table of $y = x^3$ (Example 1) find $y(1.5)$.

$x_0 = 1$, $h = 1$, $p = (1.5 - 1)/1 = 0.5$. $y_0 = 1$, $\Delta y_0 = 7$, $\Delta^2 y_0 = 12$, $\Delta^3 y_0 = 6$.

$y(1.5) = 1 + (0.5)(7) + [(0.5)(-0.5)/2](12) + [(0.5)(-0.5)(-1.5)/6](6)$

$= 1 + 3.5 - 1.5 + 0.375 =$ **3.375** (exact value $1.5^3 = 3.375$ ✓)

4. Newton's Backward Interpolation Formula

Used when x is **near the end** of the table. Put **$p = (x - x_n)/h$** (p is negative).

$$y(x) = y_n + p\nabla y_n + [p(p+1)/2!]\,\nabla^2 y_n + [p(p+1)(p+2)/3!]\,\nabla^3 y_n + \dots$$

Example 3

Find $y(4.5)$ from the same table.

$x_n = 5$, $h = 1$, $p = (4.5 - 5)/1 = -0.5$. Last row values : $y_n = 125$, $\nabla y_n = 61$, $\nabla^2 y_n = 24$, $\nabla^3 y_n = 6$.

$y(4.5) = 125 + (-0.5)(61) + [(-0.5)(0.5)/2](24) + [(-0.5)(0.5)(1.5)/6](6)$

$= 125 - 30.5 - 3 - 0.375 = \mathbf{91.125}$ (exact $4.5^3 = 91.125$ ✓)

Rule : x near start → Newton's forward ; x near end → Newton's backward ; x in the middle → central difference formulae (Gauss, Stirling, Bessel).

Example 4 (population problem)

Table of population of a town (in thousands) :

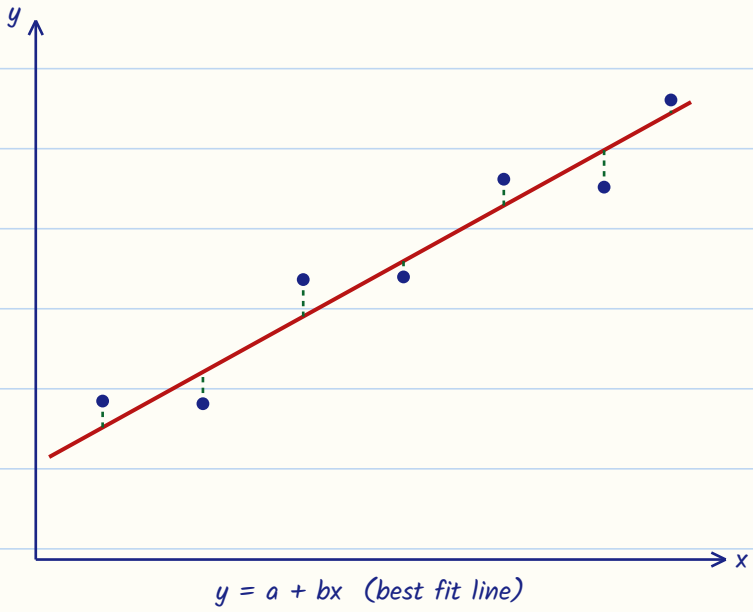
Year	1931	1941	1951	1961	1971
Population	40	60	79	103	133

Differences : $\Delta = 20, 19, 24, 30$; $\Delta^2 = -1, 5, 6$; $\Delta^3 = 6, 1$; $\Delta^4 = -5$.

Estimate the population in 1936 : $p = (1936 - 1931)/10 = 0.5$

$y = 40 + 0.5(20) + [0.5(-0.5)/2](-1) + [0.5(-0.5)(-1.5)/6](6) + [0.5(-0.5)(-1.5)(-2.5)/24](-5)$

$= 40 + 10 + 0.125 + 0.375 + 0.1953 \approx \mathbf{50.70}$ **thousand**



Principle of least squares:
minimise $E = \Sigma\, d_i^2$
where $d_i = y_i - (a + bx_i)$
(green dashed = residuals)
sum of squares of errors should be the least

5. Lagrange's Interpolation Formula

For **unequally spaced** x values (also works for equal spacing). Given (x_0,y_0), (x_1,y_1) ... (x_n,y_n) :

$$y(x) = \Sigma\, y_i \cdot L_i(x)\ ,\ L_i(x) = \Pi\ (x - x_j)/(x_i - x_j)\ (j \neq i)$$

For three points (x_0, x_1, x_2):

$$y = (x-x_1)(x-x_2)/[(x_0-x_1)(x_0-x_2)]\ y_0 + (x-x_0)(x-x_2)/[(x_1-x_0)(x_1-x_2)]\ y_1 + (x-x_0)(x-x_1)/[(x_2-x_0)(x_2-x_1)]\ y_2$$

Properties

- Polynomial has degree n for (n + 1) points and is unique.
- $L_i(x_i) = 1$ and $L_i(x_j) = 0$ for $i \neq j$.
- No need of difference table, but adding a new point means recomputing everything.

Example 5

Using Lagrange's formula, find y(3) given (0, 2), (1, 3), (2, 12), (5, 147).

$L_0 = (3-1)(3-2)(3-5) / [(0-1)(0-2)(0-5)] = (2)(1)(-2)/(-10) = 0.4$

$L_1 = (3-0)(3-2)(3-5) / [(1-0)(1-2)(1-5)] = (3)(1)(-2)/(4) = -1.5$

$L_2 = (3-0)(3-1)(3-5) / [(2-0)(2-1)(2-5)] = (3)(2)(-2)/(-6) = 2$

$L_3 = (3-0)(3-1)(3-2) / [(5-0)(5-1)(5-2)] = 6/60 = 0.1$

$y(3) = 2(0.4) + 3(-1.5) + 12(2) + 147(0.1) = 0.8 - 4.5 + 24 + 14.7 = \mathbf{35}$

The polynomial is $y = x^3 + x^2 - x + 2$, check : $27 + 9 - 3 + 2 = 35$ ✓

Inverse interpolation

To find x for a given y, interchange the roles : apply Lagrange's formula with x and y swapped, $x = \Sigma\, x_i \cdot \Pi\, (y - y_j)/(y_i - y_j)$.

Newton vs Lagrange

Newton forward / backward	Lagrange
Equal intervals only	Equal or unequal intervals
Uses difference table	Direct, no table
Easy to add more terms	Must redo whole work
Less calculation	More calculation

6. Curve Fitting : Method of Least Squares

Given data (x_i, y_i), $i = 1 \ldots n$, we find a curve $y = f(x)$ that **best fits** the data (need not pass through every point).

Principle of least squares : choose the constants so that the sum of squares of errors $E = \Sigma\, [\, y_i - f(x_i)\,]^2$ is minimum. (Put $\partial E/\partial a = 0$, $\partial E/\partial b = 0$ to get **normal equations**.)

6.1 Straight line $y = a + bx$

Normal equations :

$$\Sigma y = n\, a + b\, \Sigma x$$

$$\Sigma xy = a\,\Sigma x + b\,\Sigma x^2$$

Solve for a and b.

Example 6

Fit a straight line to : x = 1,2,3,4,5 ; y = 14,27,40,55,68.

x	y	xy	x^2
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
15	204	748	55

204 = 5a + 15b ; 748 = 15a + 55b. Multiply first by 3 : 612 = 15a + 45b. Subtract : 136 = 10b → **b = 13.6**, a = (204 − 204)/15 = 0.

So the line is **y = 13.6 x**.

6.2 Second degree parabola $y = a + bx + cx^2$

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

Example 7

Fit $y = a + bx + cx^2$ to : x = 0,1,2,3,4 ; y = 1, 1.8, 1.3, 2.5, 6.3.

Sums : n = 5, $\Sigma x = 10$, $\Sigma x^2 = 30$, $\Sigma x^3 = 100$, $\Sigma x^4 = 354$, $\Sigma y = 12.9$, $\Sigma xy = 37.1$, $\Sigma x^2y = 130.3$.

12.9 = 5a + 10b + 30c ; 37.1 = 10a + 30b + 100c ; 130.3 = 30a + 100b + 354c

Solving : **a = 1.42 , b = −1.07 , c = 0.55**. So $y = 1.42 - 1.07x + 0.55x^2$.

6.3 Exponential curve $y = a\,e^{bx}$

Take logs (base 10) : $\log y = \log a + (b \log e)\,x$, i.e. **$Y = A + Bx$**

where $Y = \log y$, $A = \log a$, $B = b \log e$. It becomes a straight line fit in (x, Y).

Normal equations : $\Sigma Y = nA + B\Sigma x$, $\Sigma xY = A\Sigma x + B\Sigma x^2$. Then $a = 10^A$, $b = B / \log e$ ($\log e = 0.4343$).

Example 8

Fit $y = a\,e^{bx}$: x = 1,2,3,4,5 ; y = 2.0, 4.5, 10.1, 22.8, 51.2.

$Y = \log y$ = 0.3010, 0.6532, 1.0043, 1.3579, 1.7093. $\Sigma x = 15$, $\Sigma x^2 = 55$, $\Sigma Y = 5.0258$, $\Sigma xY = 18.5985$.

$5.0258 = 5A + 15B$; $18.5985 = 15A + 55B \rightarrow$ **B = 0.3521 , A = −0.0512**

a = 10^(−0.0512) = 0.889 , b = 0.3521/0.4343 = 0.811. Curve : **y = 0.889 e^(0.811 x)**

Similarly $y = a\,x^b$ (power curve) ...

Take log : $\log y = \log a + b \log x$, so $Y = A + bX$ with $X = \log x$, $Y = \log y$. Fit a line in (X, Y).

6A. More Solved Examples

Example 9 : Newton forward with non-polynomial data

Given sin 45° = 0.7071, sin 50° = 0.7660, sin 55° = 0.8192, sin 60° = 0.8660. Find sin 52°.

x°	y = sin x	Δy	Δ²y	Δ³y
45	0.7071	0.0589	−0.0057	−0.0007
50	0.7660	0.0532	−0.0064	
55	0.8192	0.0468		
60	0.8660			

$x_0 = 45$, $h = 5$, $p = (52 - 45)/5 = 1.4$.

$y = 0.7071 + 1.4(0.0589) + [1.4(0.4)/2](-0.0057) + [1.4(0.4)(-0.6)/6](-0.0007)$

$= 0.7071 + 0.08246 - 0.00160 + 0.00004 =$ **0.7880** (actual sin 52° = 0.7880 ✓)

Example 10 : finding the missing value

Find the missing term : x = 0,1,2,3,4 ; y = 1, 3, 9, –, 81.

Four values are known so $\Delta^4 y_0 = 0$: $y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$

$81 - 4y_3 + 54 - 12 + 1 = 0 \rightarrow 124 = 4y_3 \rightarrow$ **$y_3 = 31$**.

Example 11 : Lagrange with three points

Find y(2) from (1, 1), (3, 27), (4, 64) (these are cubes).

$y(2) = [(2-3)(2-4)]/[(1-3)(1-4)] \cdot 1 + [(2-1)(2-4)]/[(3-1)(3-4)] \cdot 27 + [(2-1)(2-3)]/[(4-1)(4-3)] \cdot 64$

$= (2/6)(1) + (-2/-2)(27) + (-1/3)(64) = 0.3333 + 27 - 21.3333 =$ **6.0**

(Cube of 2 is 8, the difference arises because we used only a quadratic through three points.)

Error in interpolation

If $y = f(x)$ is approximated by a polynomial $P_n(x)$ through $(n + 1)$ points then

$$f(x) - P_n(x) = [\,(x-x_0)(x-x_1)\ldots(x-x_n) / (n+1)!\,] \cdot f^{(n+1)}(\xi)$$

where ξ is some point with $x_0 < \xi < x_n$.

Error is small when x is near the middle of the data points and h is small.

Example 12 : straight line with intercept

Fit $y = a + bx$: $x = 0, 1, 2, 3, 4$; $y = 1, 3, 4, 6, 8\ldots$ $(n = 5)$

$\Sigma x = 10$, $\Sigma y = 22$, $\Sigma xy = 0 + 3 + 8 + 18 + 32 = 61$, $\Sigma x^2 = 30$.

$22 = 5a + 10b$; $61 = 10a + 30b$. From first : $44 = 10a + 20b$.

Subtract : $17 = 10b \rightarrow b = 1.7$, then $a = (22 - 17)/5 = 1.0$.

Best line : **$y = 1 + 1.7\,x$**. Estimate at $x = 5$: $y = 9.5$.

Example 13 : power curve $y = a\,x^b$

$x = 1, 2, 3, 4$; $y = 2.0, 5.7, 10.4, 16.0$ (roughly $y = 2x^{1.5}$).

$X = \log x = 0, 0.3010, 0.4771, 0.6021$; $Y = \log y = 0.3010, 0.7559, 1.0170, 1.2041$.

$\Sigma X = 1.3802$, $\Sigma Y = 3.2780$, $\Sigma XY = 1.4377$ (approx), $\Sigma X^2 = 0.6807$ (approx)

Normal equations on $Y = A + bX$ give $b \approx 1.5$, $A \approx 0.30 \rightarrow a = 10^{0.30} = 2.0$. So $y \approx 2.0\,x^{1.5}$.

Remarks on curve fitting

- Number of normal equations = number of constants in the curve.
- To reduce the size of calculation, shift origin : $u = (x - A)/h$ when x values are equally spaced.
- Goodness of fit : compare Σ of squared residuals of different curves, least value is best.

7. Quick Revision

- $\Delta = E - 1$, $\nabla = 1 - E^{-1}$, $p = (x - x_0)/h$ for forward, $p = (x - x_n)/h$ for backward.
- Forward : near top of table ; Backward : near bottom ; Lagrange : unequal gaps.
- Least squares : straight line (2 normal eqs), parabola (3 normal eqs), exponential (take log first).

Exam questions

- Derive Newton's forward interpolation formula.
- Using Lagrange's formula, find y(10) given (5,12), (6,13), (9,14), (11,16) → y(10) = 14.67.
- Fit $y = ab^x$ or $y = a\,e^{(bx)}$ to the given data by least squares.
- Prove that $\Delta = E - 1$ and $\nabla = 1 - E^{-1}$; find the missing term in a table.